

Chapter 4. Market Phase Analysis

Lecture notes | First semester

Benjamin C. Sotelo | Institute of Accounts, Business and Finance, FEU Manila

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What this chapter is about

These notes are the record of what Chapter 4 covered, written to be read on their own. If you were in the room, they are what to revise from. If you missed the session, they are the session. They follow the lecture in the same order. The nine sections are the book's own sections 4.1 to 4.9, so Section 3 of these notes is the book's section 4.3, and the book's 4.10, its summary, is the summary at the back.

The chapter asks one question nine ways: which phase is the market in? Read it with the pictures beside you. A figure is the book's. A chart is one this course drew, on invented prices, where the book makes a point in words and draws nothing. Every term is defined once, in the same words as the slides, and listed again at the back. The check yourself boxes are not assessed.

Six of the nine sections borrow a tool that a later chapter of the book teaches. Where that happens, or where the book says two different things, these notes say so rather than filling the gap from somewhere else, and Section 9.3 collects every one of those places.

After studying this chapter you should be able to:

1. Understand the structural and functional characteristics of the three market phases.
2. Forecast market phase accurately via its behavioral and cyclical properties.
3. Identify potential market regime changes via volume action.
4. Describe the inertial and momentum characteristics for each market phase.
5. Identify specific price pattern formations associated with each phase.
6. Compare and contrast Eastern and Western approaches to interpreting market phases.

Chapter scope follows Lim, M. (2016), *The Handbook of Technical Analysis*, chapter 4. Figures are reproduced from that text and remain the publisher's copyright. Charts are drawn for this course on invented prices.

Section 1

Dow Theory of Market Phase

The two basic phases and the two kinds of consolidation, why the kind is known only afterwards, who buys and who sells in each phase, the signs of each consolidation, and when a consolidation is over. This is the book's section 4.1.

1.1 Markets move in phases

Chapter 2 gave Dow Theory's tenet that a primary trend has three phases: accumulation, trending and distribution. That description is now regarded as the basic underlying characteristic of market action, and this chapter is about nothing else. Phase analysis applies across all time frames. **The book puts the stakes plainly: not recognizing the phases and their transitions greatly disadvantages the practitioner.**

One question runs through all nine sections: which phase is the market in? Each section answers it with a different kind of evidence.

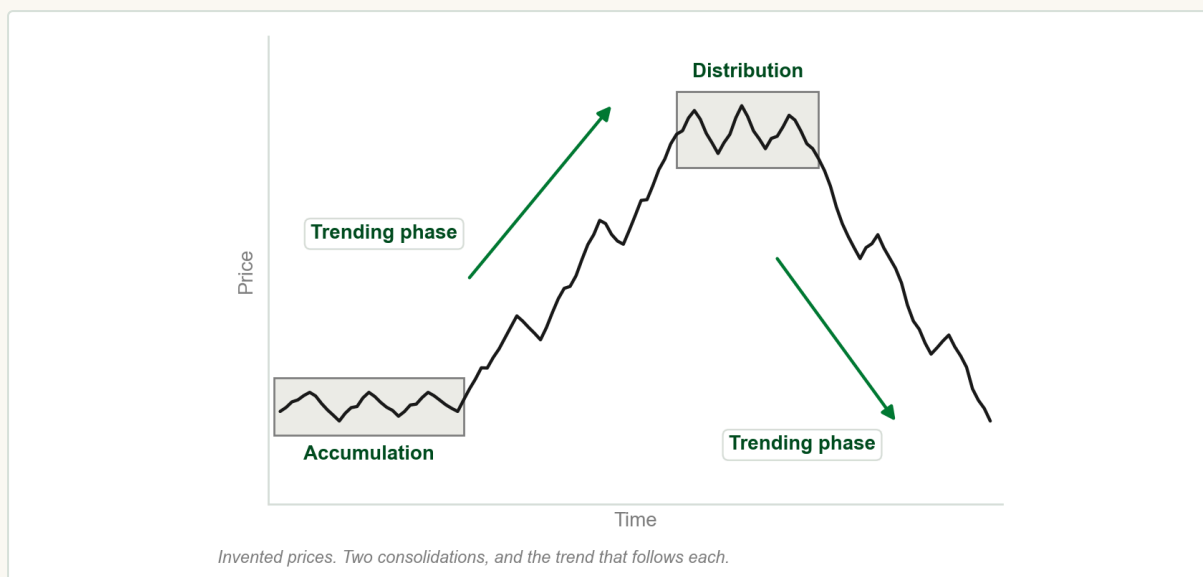


Chart A One price line through the phases, in order.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart A draws the whole idea on one invented price line: an accumulation, the trend up out of it, a distribution, and the trend down out of that. Read it from left to right and name each stretch before you read its label.

Three phases, but only two kinds. Either the market is going somewhere or it is not, and the sideways stretch is the one that needs a definition. Picture a share that swings between 40 and 44 pesos for four months and goes nowhere. It is consolidating: price ranges, or makes small corrective moves, instead of trending.

Consolidation phase. One of only two basic phases, the other being the trending phase. A market consolidates, when expressing indecision, rest or exhaustion, by ranging or by making relatively small corrective moves. It subdivides into accumulation and distribution.

The book's first review question asks for that definition, and the book never sets one apart in a single place. **The wording above joins two of the book's own sentences: the one in its section 4.1 that there are only two basic phases, and the one in its section 4.2 on how a market consolidates.**

A tree: market phase splits into consolidation and trend, and consolidation splits again into accumulation and distribution.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.1 The phases as a tree.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.1 is the same statement drawn as a tree.

Read it from the top down: market phase splits into consolidation and trend, and consolidation splits again into accumulation and distribution.

1.2 Accumulation or distribution is known only afterwards

If the market rises after consolidating, the consolidation was accumulation, which is buying activity. If it declines after consolidating, it was distribution, which is selling activity. **Which one it is can only be ascertained after the fact.**

While the range is forming, the two look alike. That is why the book makes it one objective of technical analysis to look for evidence of which is taking place. The rest of the chapter is that hunt: volume, chart patterns, moving averages, momentum, sentiment and cycles.

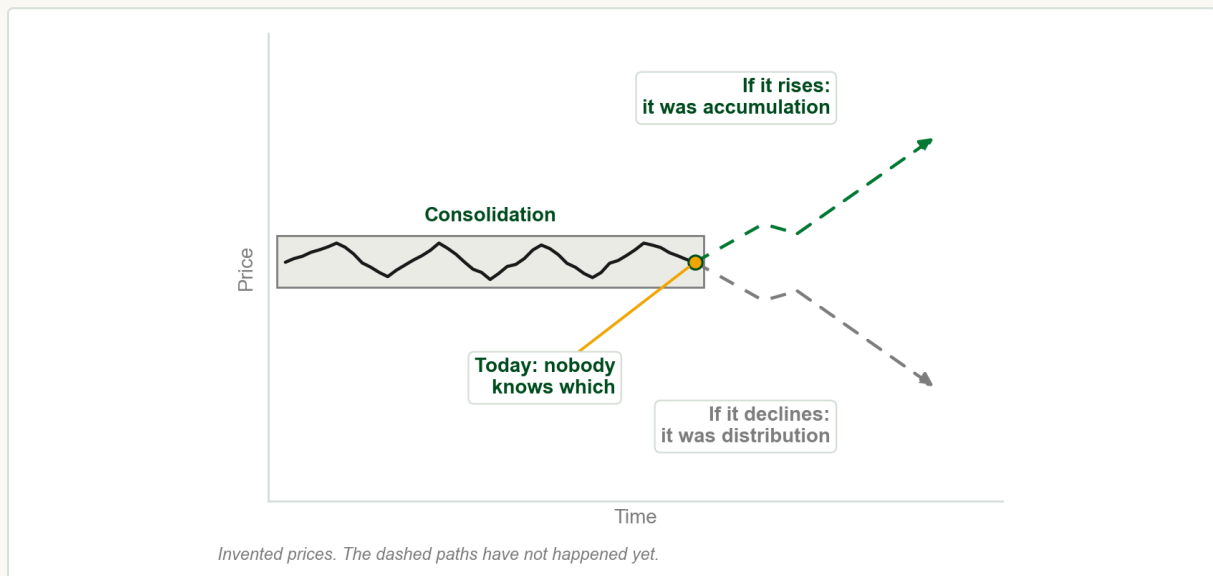


Chart B One consolidation, two outcomes.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart B shows one consolidation with two dashed paths leaving it. Cover the two paths with your hand. The range is the same either way, and the name it earns depends on the move that comes next.

1.3 The first evidence: volume in a consolidation

Volume normally declines gradually during a consolidation. Very low volume is a strong sign that the consolidation may be ending and a trend starting. On a daily chart a consolidation normally lasts about three to six months, or longer.

Figure 4.2 is the daily Dow Jones Industrial Average with its accumulation, trending and distribution phases boxed and named. Look under each box, in the volume panel, where the decline in volume through that consolidation is drawn in. Then look at the last box, at the top right. **It is labelled only consolidation, because what followed it was not yet known.**

1.4 The same phases at every time scale

A primary trend typically spans months to years in the equity markets. In commodity futures it tends to be shorter, because high leverage induces greater volatility. Dow described the equity markets, but the phases are found across all markets. They show on sub-hourly charts too. The book calls that micro phase action, and says it is shaped by the macro phase behavior around it.

The daily Dow Jones Industrial Average with its accumulation, trending and distribution phases boxed and named, a last box named only consolidation, and a volume panel below with the decline in volume through each consolidation drawn in.

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Figure 4.2 The phases, and their volume, on the daily Dow.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

A 15 minute EURUSD chart with distribution, trend and accumulation phases boxed one after another, titled micro phase action.

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Figure 4.3 Micro phase action on a 15 minute chart.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.3 is a 15 minute EURUSD chart, and it has a distribution, a trend and an accumulation boxed one after another, just as the daily chart of the Dow did. Looking at more than one time frame is how to see both at once: the macro phase first, then the micro phases inside it.

CHECK YOURSELF

A share has ranged between 40 and 44 pesos for four months. Can you say today whether that is accumulation or distribution? What would you look for while you wait?

1.5 Who buys and who sells in each phase

An accumulation normally follows a deep and rapid decline, on very negative data and bearish headlines. The uninformed are extremely bearish and sell at whatever price is available. The informed buy from them, which is a contrarian approach and needs deep pockets. They gather shares very gradually, careful not to drive prices up too fast. **The sell-off that ends the downtrend is called a selling climax.** The book describes the informed buying into it as very large responsive buying, and that buying is what creates the final and decisive bottom.

Chart C draws it: a deep, rapid decline ends in a selling climax, where the uninformed sell and the informed buy, and a long accumulation range follows through which the informed keep gathering shares.

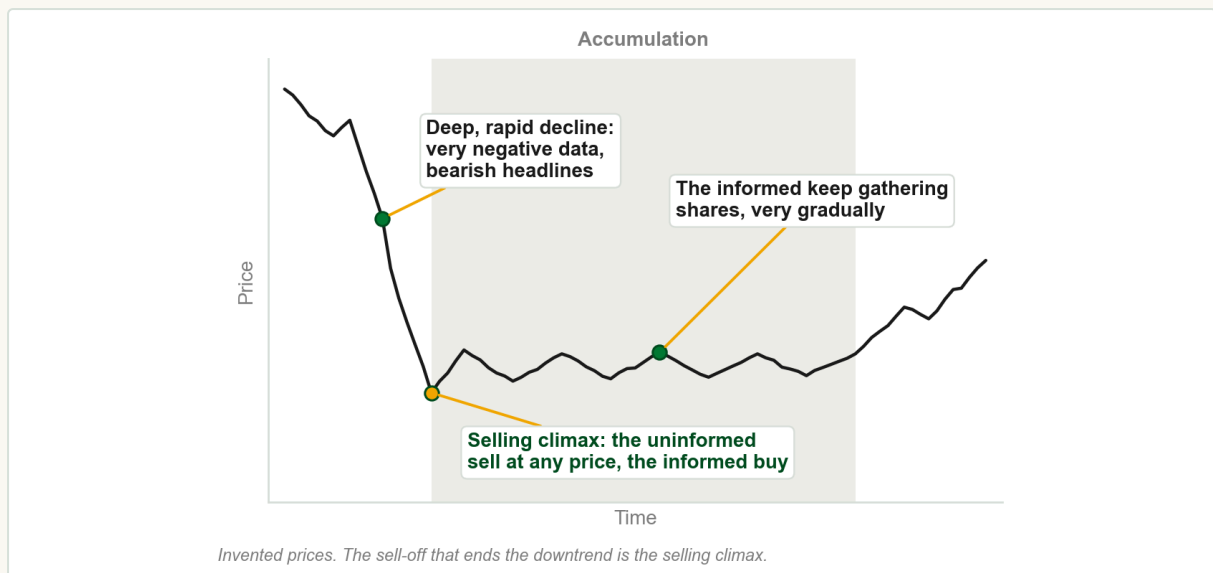


Chart C Accumulation: the selling climax, and after.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

The uptrend that follows feeds on itself. Technical traders get in early, on a clear upside breakout, and market savvy investors follow. As the uptrend becomes obvious the public is drawn in, and the bears cover their shorts. Then comes regret bias: those who missed out, or sold too early, buy at every dip. At even higher prices the herd uses more margin, even borrowing to invest. **The book calls this a vicious positive feedback cycle.**

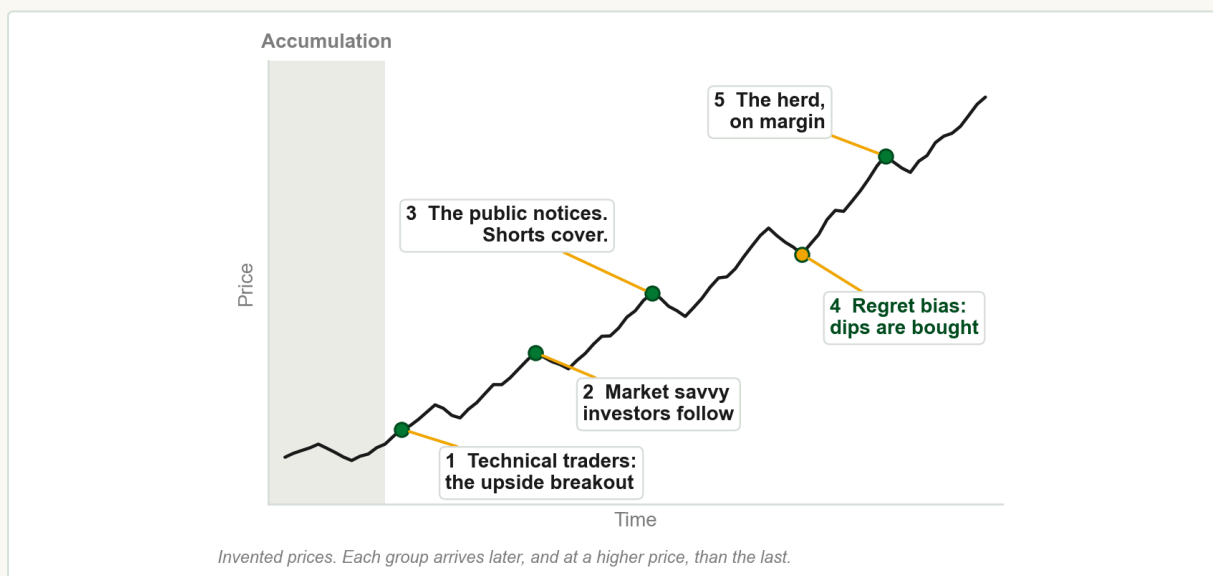


Chart D The uptrend: who joins, in the order they join.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart D marks the five groups in the order they arrive, numbered 1 to 5. Each group arrives later, and at a higher price, than the one before it.

The downtrend is the same cycle running downward. It begins with a breakdown from a distribution range, on increasingly bearish data. The uninformed begin to unload, and previously bullish participants liquidate. Past the threshold of pain and the margin limits, capital flows out of the market. Bearish sentiment intensifies as prices sink. Further unexpected declines attract more public attention, and that causes the larger liquidation.

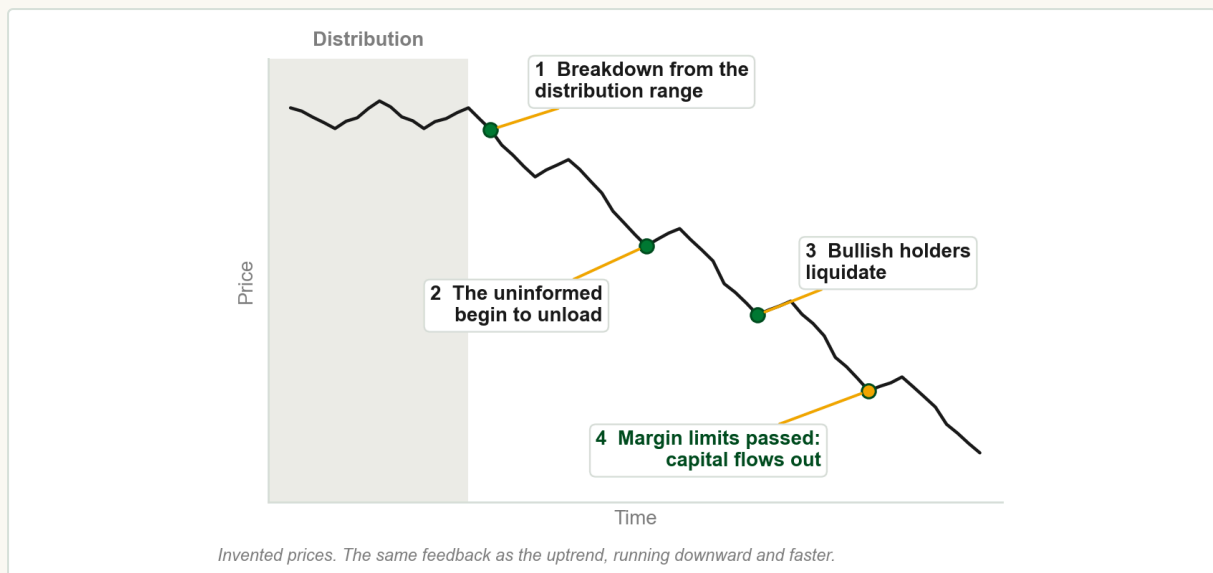


Chart E The downtrend: the same feedback, downward.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart E is Chart D turned over, marked in order: the breakdown, the uninformed beginning to unload, bullish holders liquidating, and capital flowing out once margin limits are passed.

A distribution is the accumulation story upside down. It normally follows a strong and rapid rise, on very positive data and the most bullish headlines. The uninformed buy at whatever price is available, a state normally referred to as irrational exuberance, and margin debt is skyrocketing. The informed sell to them very gradually, careful not to drive prices down too rapidly. **The surge that ends the uptrend is called a blow-off, or buying climax.**

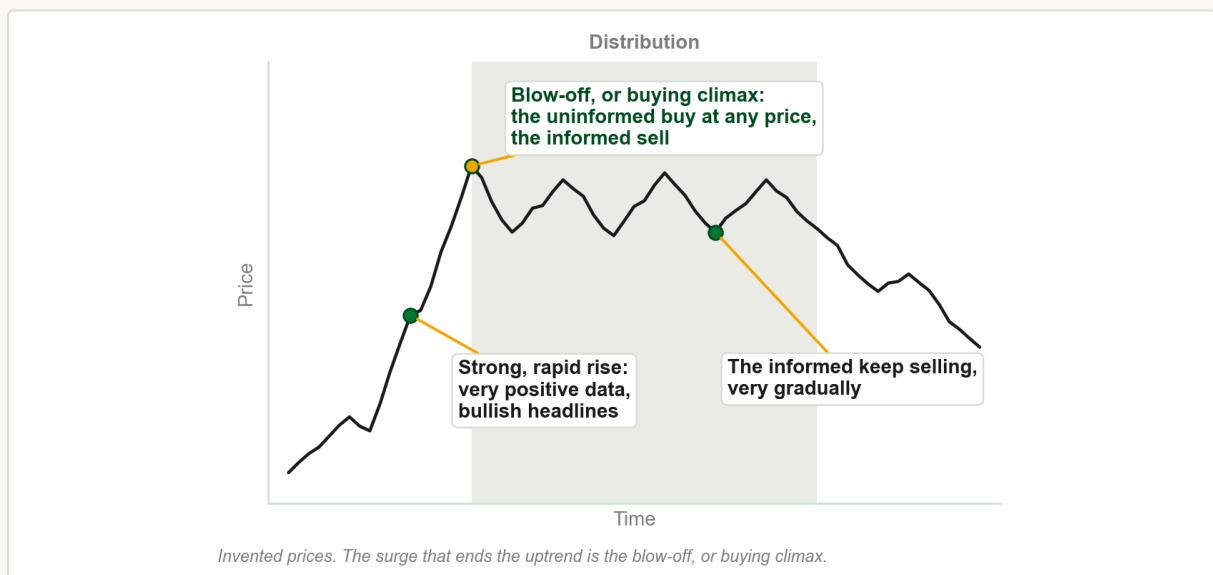


Chart F Distribution: the blow-off, and after.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart F is the mirror of Chart C. A strong, rapid rise ends in a blow-off, where the uninformed buy and the informed sell, and a distribution range follows through which the informed keep selling. You should be able to write the accumulation paragraph from the distribution one, line by line, and the other way round.

1.6 Why tops are shorter and rougher than bottoms

The book makes three comparisons. Accumulation normally lasts longer than distribution. The uptrend tends to be more prolonged than the downtrend. Distributions tend to be much more volatile than accumulations. **One reason covers all three: at higher prices more capital, and more unrealized profit, is at risk.** At the bottom, lower prices mean a lower amount of capital at risk, so nobody is in a hurry. At the top everyone has profit to protect.

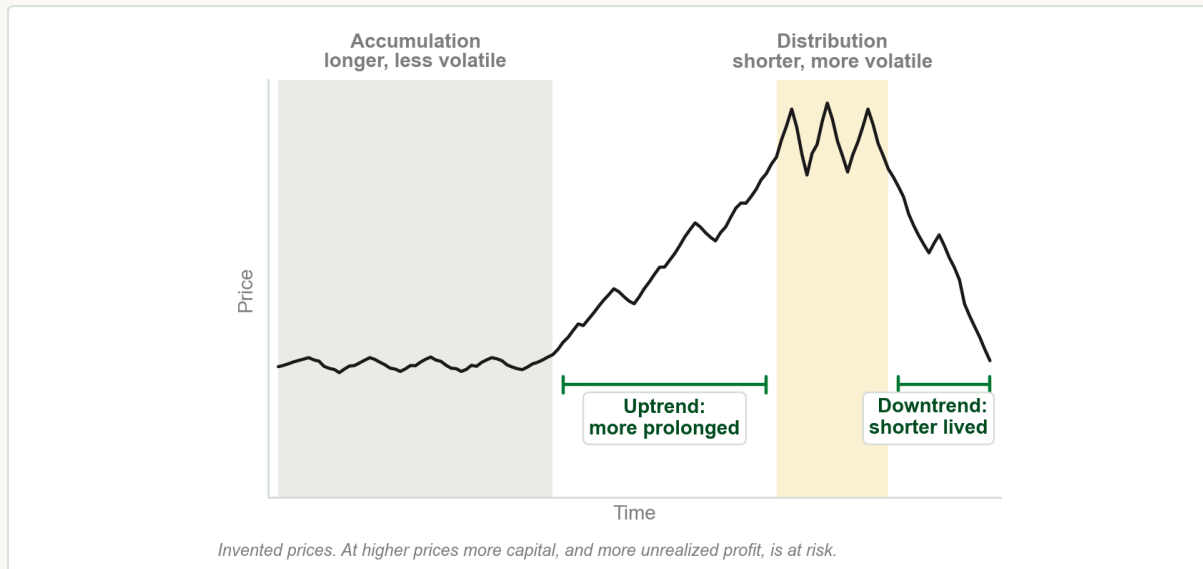


Chart G The three comparisons on one cycle.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart G draws one cycle to those three comparisons: a long, quiet accumulation, a prolonged uptrend, a short and volatile distribution, and a short downtrend.

1.7 The signs of each consolidation

The label is known only afterwards, but the book lists what to look for while a consolidation is still forming. The two lists mirror each other, and the second adds one item: signs of exhaustion.

The signs of an accumulation

- It comes after a relatively rapid decline, ideally near a significant prior bottom.
- There is no evidence of lower troughs being formed.
- Volume begins to subside as a potential upside breakout approaches.
- The longer it lasts, the more powerful the breakout, typically on a surge in volume.
- It starts a new primary bull market, or a shorter-term uptrend.

The signs of a distribution

- It comes after a prolonged uptrend, ideally near a significant prior top.
- Signs of market exhaustion show, and no higher peaks form.
- Volume begins to subside as a potential downside breakout approaches.
- The longer it lasts, the more powerful the breakout, typically on a surge in volume.
- It starts a new primary bear market, or a shorter-term downtrend.

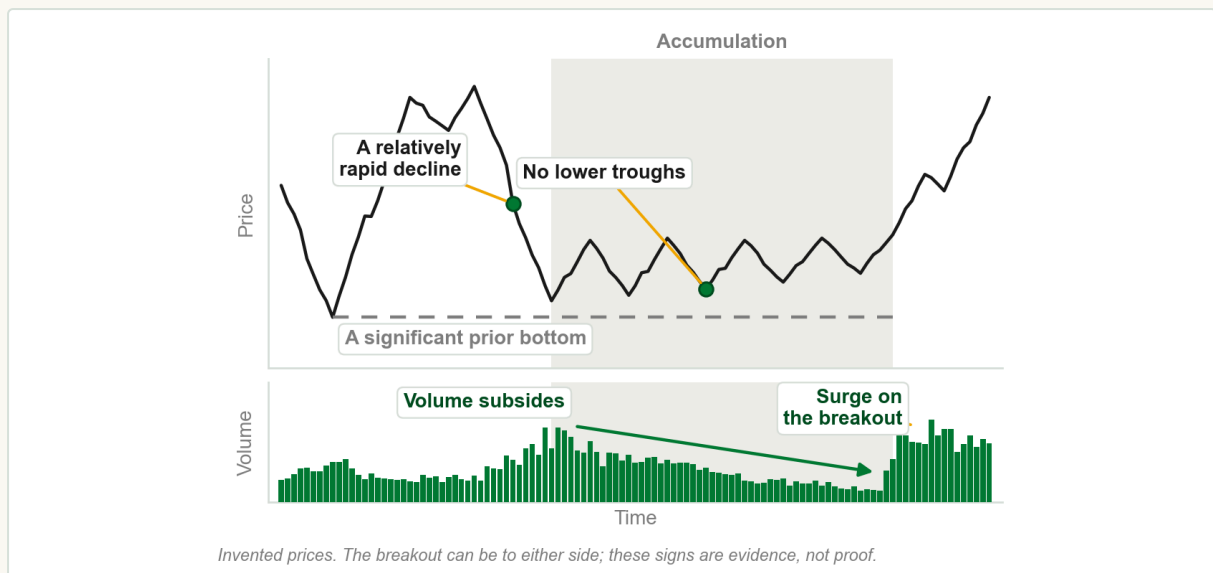


Chart H The signs of an accumulation, over volume.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

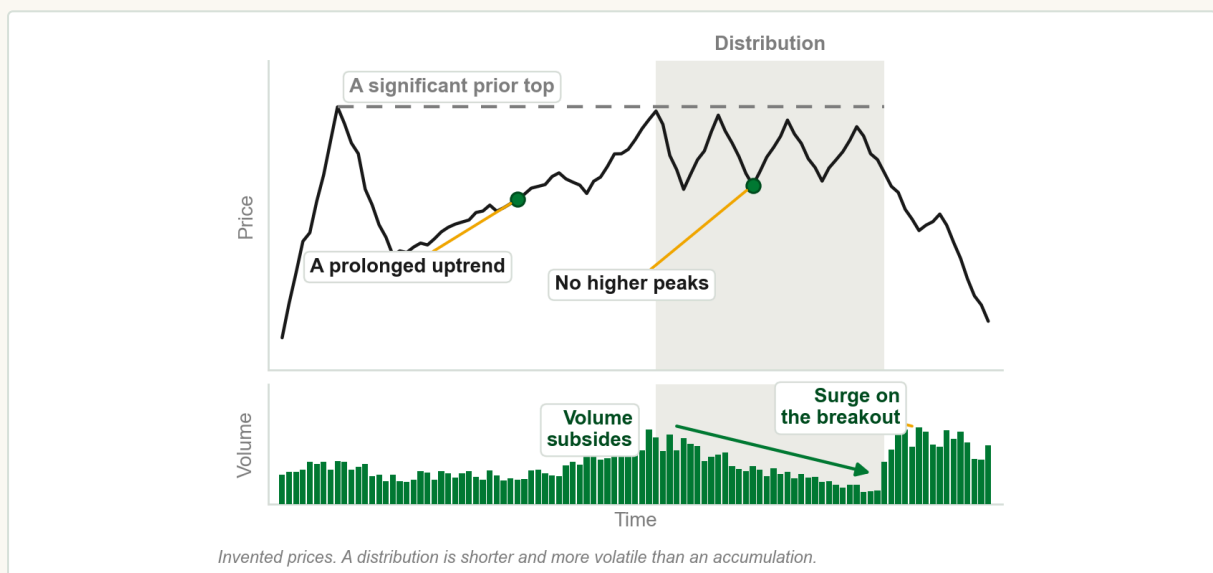


Chart I The signs of a distribution, over volume.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart H draws the first list: an accumulation after a rapid decline, sitting on the level of a significant prior bottom with no lower troughs, over a volume panel that subsides through the range and surges on the breakout. Chart I is its mirror, under the level of a significant prior top with no higher peaks.

The first list is the answer to the review question that asks how you would determine if accumulation is taking place. **Treat both lists as evidence and not as proof.** The book says the breakout can be to either side of the range.

1.8 When is a consolidation over?

There is no absolute consensus on the exact point where a consolidation ends. Some practitioners look for a 3 to 5 percent rise above the highest peak of the range. Some wait for the minimum measuring objective to be met. **Most look for a clear and simple technical breakout from the range.** Even then, where the breakout occurs depends on the price filter used. Once a consolidation is complete, the book adds, it follows logically that a trend may already be in effect.

A consolidation range with the three popular completion levels marked above it: a technical breakout, a 3 to 5 percent rise above the highest peak, and a one to one minimum target projection of the range.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.4 Three popular completion levels.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.4 marks the three levels above one consolidation range: a technical breakout, a 3 to 5 percent rise above the highest peak, and a one to one minimum target projection of the range. That last label is the minimum measuring objective under another name.

The minimum measuring objective is the least a breakout is expected to travel, and it is usually the height of the range it broke out of. Take the share that ranges between 40 and 44 pesos. The range is 4 pesos high, so on an upside breakout the minimum target is $44 + 4 = 48$ pesos.

Minimum measuring objective. The expected minimum price target, which is usually a price excursion equal to the range or height of the consolidation.

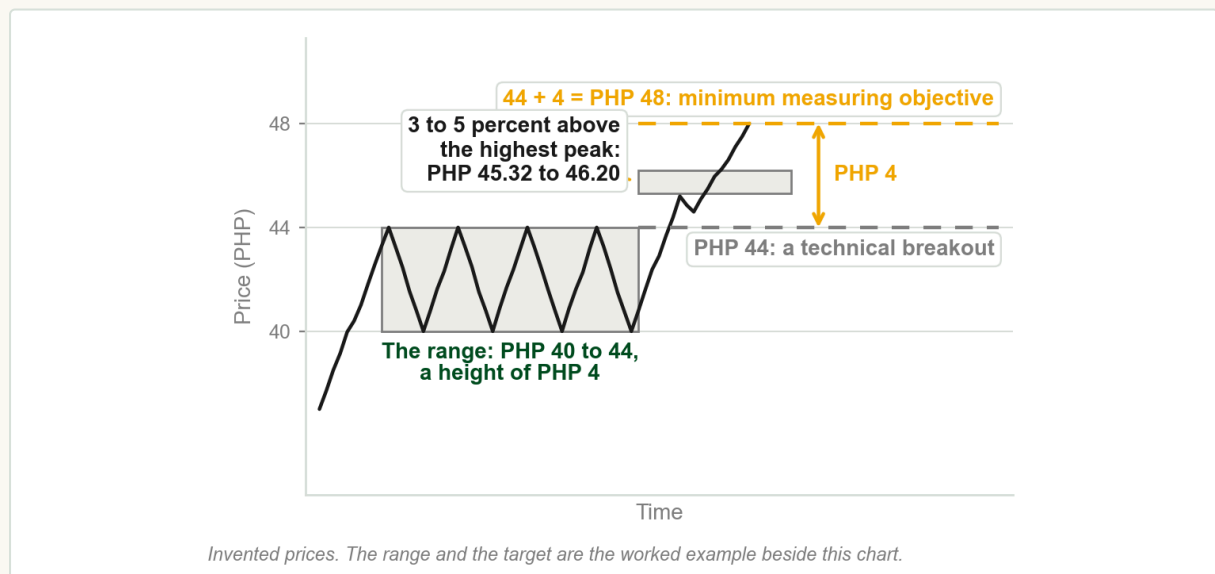


Chart J The three completion levels, in pesos.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart J puts all three levels on that same range, in pesos: the breakout at 44, the band 3 to 5 percent above the peak at 45.32 to 46.20, and the minimum measuring objective at 48.

CHECK YOURSELF

A share ranges between 30 and 35 pesos and then breaks out upward. Work out its minimum measuring objective, and the band that is 3 to 5 percent above the highest peak.

Section 2

Chart Pattern Interpretation of Market Phase

Reversal and continuation patterns, what makes a pattern a consolidation pattern, the bias a pattern carries in itself and the bias it takes from where it sits, and why agreement between the two is what to look for. This is the book's section 4.2.

2.1 Chart patterns show the phase

Market phase can also be identified by the type of chart pattern found at each stage. Most chart patterns belong to one of two groups, reversal or continuation. Accumulation and distribution tend to hold reversal patterns, and continuation patterns form within the trend. **So the kind of pattern on the chart is evidence of the phase the market is in.**

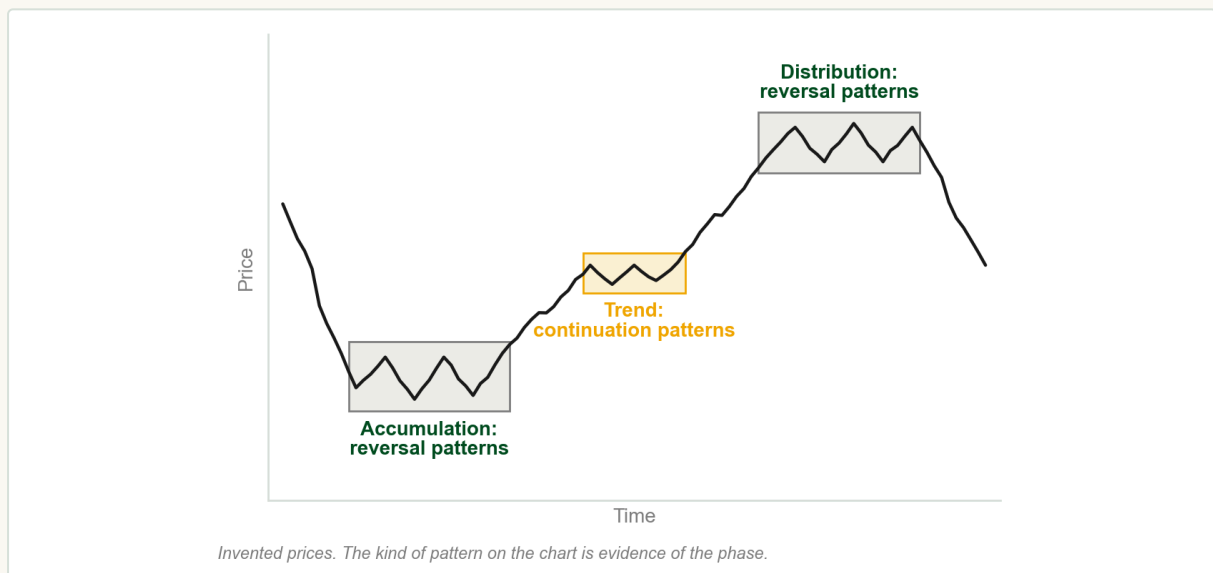


Chart K Which kind of pattern sits in which phase.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart K boxes three patterns on one price line: a reversal pattern at the bottom as accumulation, a continuation pattern inside the trend, and a reversal pattern at the top as distribution. Reversal and continuation are the two words to hold on to, because every pattern in this section is filed under one of them.

2.2 Consolidation patterns, and the one exception

A pattern that stays inside a range and is not trending anywhere is a consolidation pattern. A share that drifts between 40 and 44 pesos for six weeks while a triangle forms is making one. The book's wording sets two tests, and both are needed: a confined range, and no significant trend component.

Consolidation pattern. A chart pattern is usually regarded as a consolidation pattern if it unfolds within a relatively confined and well defined price range, and lacks any significant trend component.

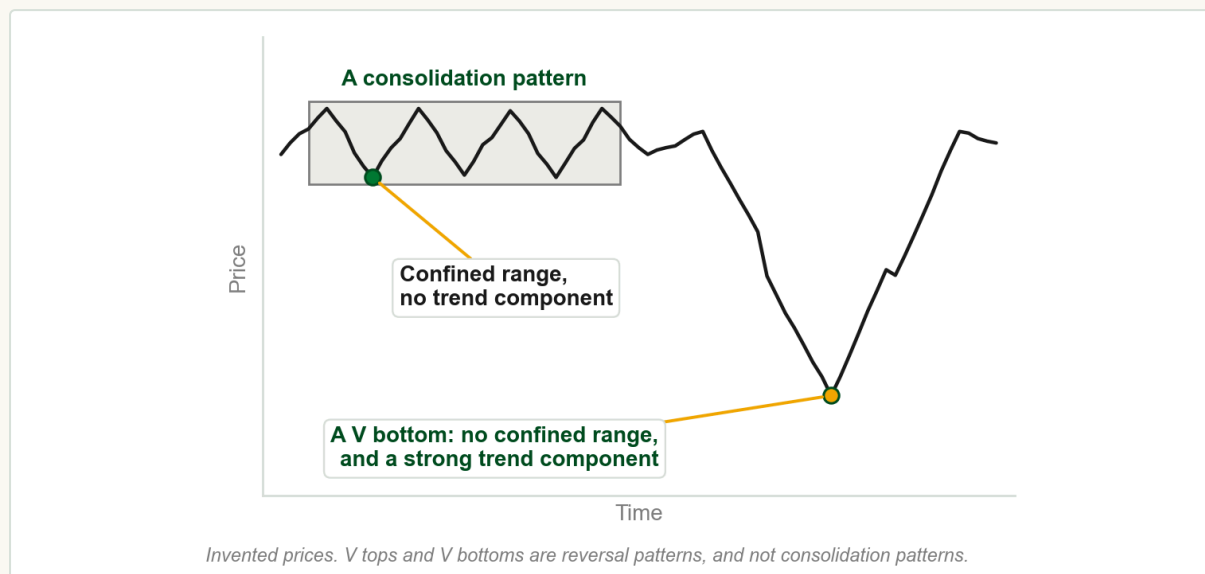


Chart L A consolidation pattern, beside a V bottom.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart L holds the two tests against two shapes. On the left a pattern is boxed inside a confined range with no trend component. On the right is a V bottom, which has no confined range and a strong trend component, and so fails both.

A market that is expressing indecision, rest or exhaustion ranges, or makes relatively small corrective moves. **That is why most reversal and continuation patterns are also consolidation patterns, and it is the answer to the review question that asks why.** V tops and V bottoms are the exception. They are the hardest reversals to forecast, seen only in hindsight, and they occur more often during distributions than accumulations.

A 4 hour USDCAD chart: a head and shoulders distribution labelled intrinsically bearish, a decline into a sharp V bottom labelled a rapid accumulation phase, and a new consolidation after the rise.

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Figure 4.8 A head and shoulders top, and a V bottom.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.8, a 4 hour USDCAD chart, has one of each. A head and shoulders forms the distribution and is labelled intrinsically bearish, a word the next subsection defines. The decline that follows ends in a sharp V bottom, labelled a rapid accumulation phase: an accumulation with no range to watch it forming in. A new consolidation follows the rise.

2.3 Intrinsic bias, and the book's three lists

Some patterns lean one way in themselves, wherever they appear. An ascending triangle is intrinsically bullish: wherever it occurs, it is inherently a bullish indication.

Intrinsic bias. The inherent bullish or bearish sentiment associated with a chart pattern that is independent of location, where sentiment is strictly based on the nature of the pattern.

A EURUSD chart with an inverted head and shoulders boxed at the bottom of a decline, labelled an intrinsically bullish pattern and a reversal pattern, and the trend phase rising out of it.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.6 An intrinsically bullish reversal pattern.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.6 shows an inverted head and shoulders boxed at the bottom of a decline on a EURUSD chart, labelled an intrinsically bullish pattern and a reversal pattern. It is the accumulation, and the trend phase kicks in after the upside breakout. The book points out that this chart cannot tell us the pattern's extrinsic bias, the subject of 2.4, because it shows no price history, cycle or barrier.

Intrinsically bullish: the book's eight

- Bullish pennants.
- Bullish flags.
- Ascending triangles.
- Inverted head and shoulders.
- Rounding bottoms.
- Cup and handles.
- Falling wedges.
- Double, triple and multiple bottoms.

The second review question asks for those eight, so learn them as a list. Wherever one of them forms, it is inherently a bullish indication. Chart M sketches all eight. Pairs help: the pennant and the flag, which both follow a pole, the two rounded ones, and the triangle and the wedge.

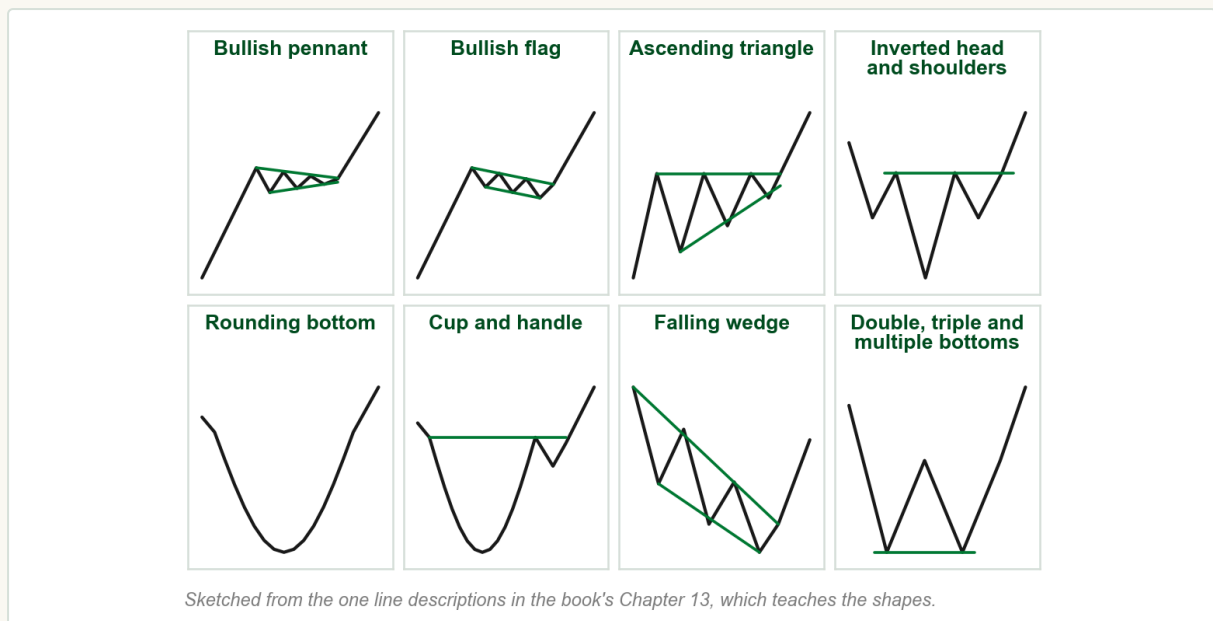


Chart M The eight intrinsically bullish patterns, sketched.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Intrinsically bearish: the book's seven

- Bearish pennants.
- Bearish flags.
- Descending triangles.
- Standard head and shoulders.
- Rounding tops.
- Rising wedges.
- Double, triple and multiple tops.

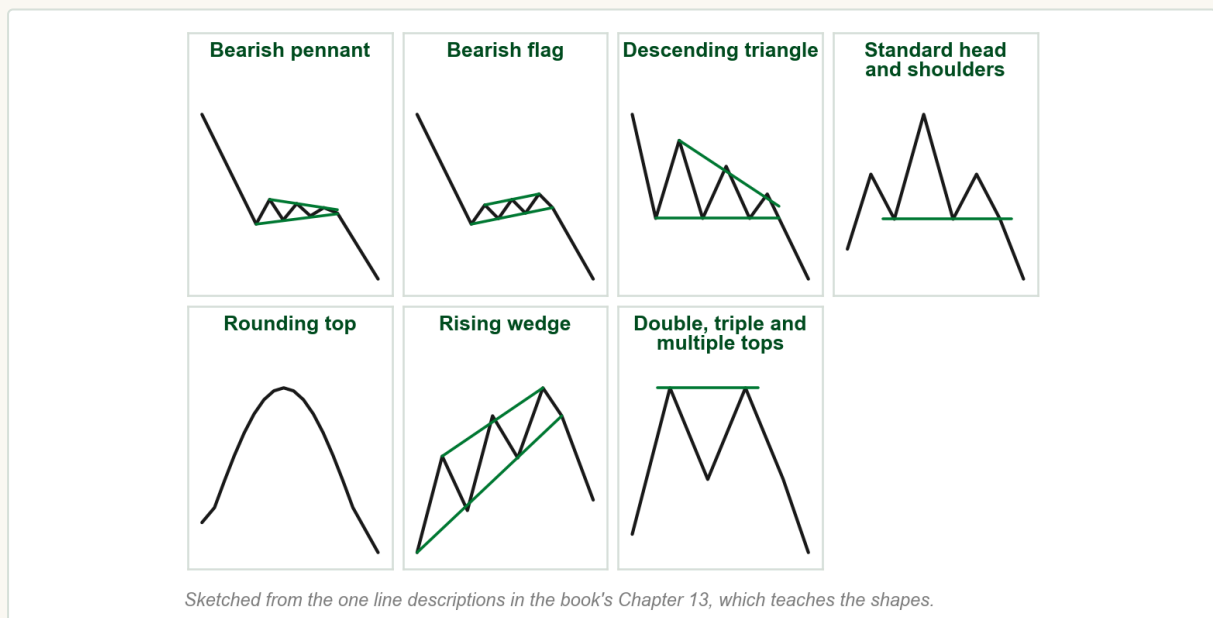


Chart N The seven intrinsically bearish patterns, sketched.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Each is a bullish pattern turned upside down, as Chart N shows. There are seven and not eight because the cup and handle is on the bullish list and has no twin on this one. Take care with the wedges, which are easy to get backwards: a falling wedge is bullish, and a rising wedge is bearish.

The third list is the neutral one. Symmetrical triangles and horizontal channels have no obvious bias in sentiment. Some authors classify them as continuation patterns, others as reversal patterns. Broadening, diamond and island formations are also neutral, but are regarded as reversal formations. **Whether a neutral pattern reverses or continues is decided beyond the pattern itself.**

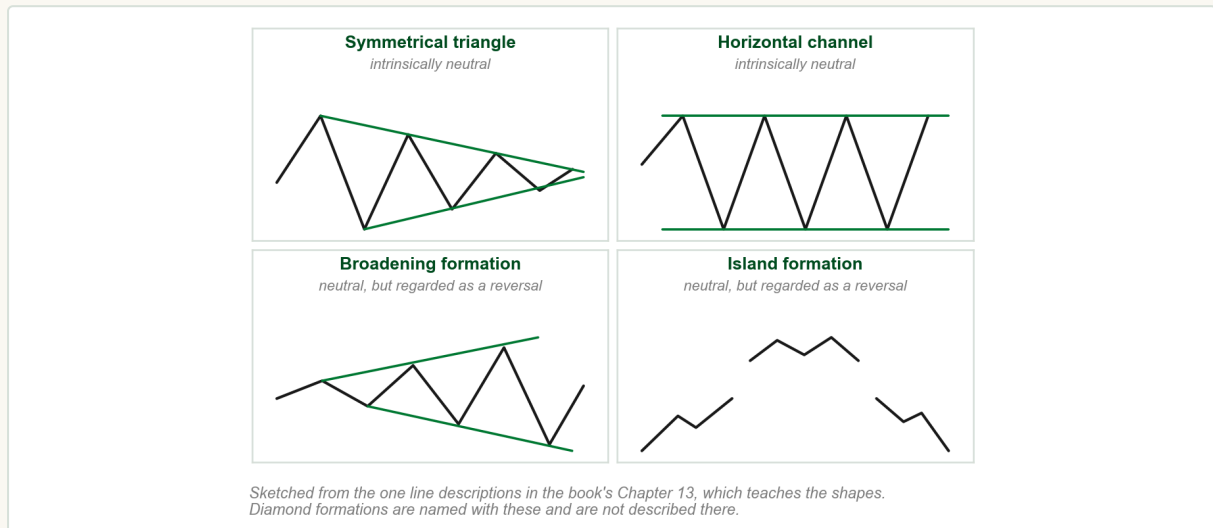


Chart O Four of the five neutral patterns, sketched.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart O sketches a symmetrical triangle and a horizontal channel, the two intrinsically neutral patterns, and a broadening formation and an island formation, which are neutral but regarded as reversals.

An honest note about the three sketches: the chapter names these seventeen patterns and describes no shape in words. It shows seven of them on real charts, in Figures 4.6 to 4.14, and the book teaches the shapes in its Chapter 13. The sketches are drawn from Chapter 13's own one line descriptions, as a preview, and from nothing outside the book. What this chapter asks of you is the lists, not the drawings. Diamond formations are named with the others and have no sketch, because the book gives no description to draw one from.

2.4 Extrinsic bias

A pattern also takes a lean from where it sits, and not from its shape.

Extrinsic bias. Location-based sentiment: sentiment that is dependent on factors that are external, or extrinsic, to the pattern.

The book's own worked example uses one shape in three places. An ascending triangle at the price level of a historically significant market bottom is intrinsically and extrinsically bullish, so the tendency for a reversal is greater. In the middle of an uptrend, a continuation is likelier. **At the price level of a historically significant market top the same triangle is extrinsically bearish, because it sits at a significant resistance.** The shape never changes and is bullish throughout. Only the location does.

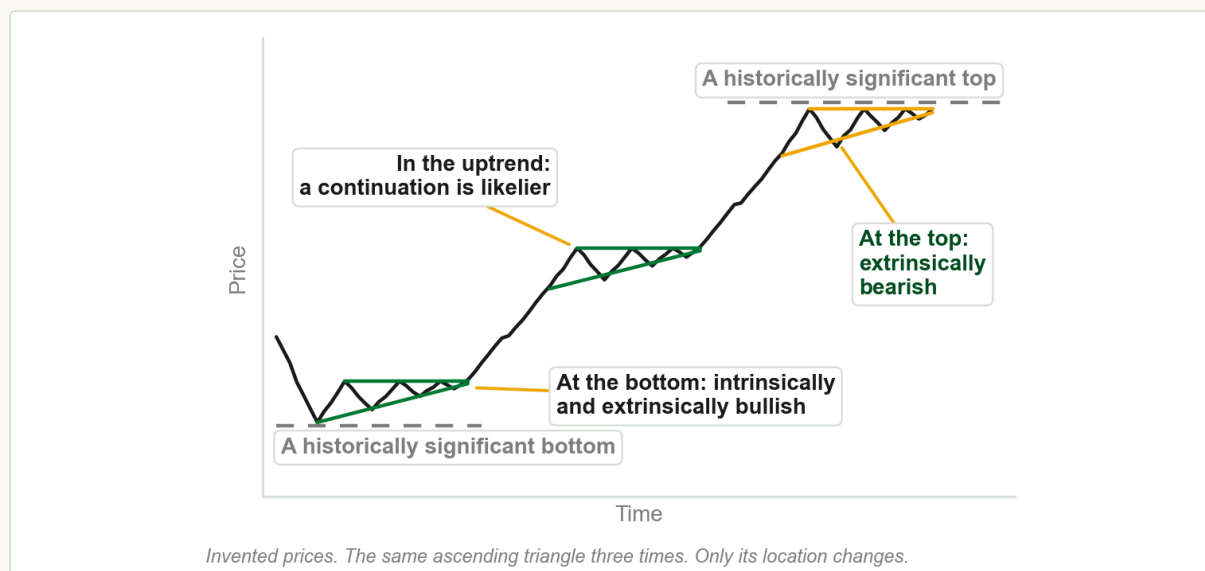


Chart P One ascending triangle in three locations.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart P draws that example: the same ascending triangle at a historically significant bottom, in the middle of an uptrend, and at a historically significant top.

What sets the extrinsic bias

1. The direction of the preceding trend.
2. Location with respect to historical extremes in price.
3. Location with respect to the phase of an underlying market cycle.
4. Location with respect to other supportive and resistive overlay barriers.
5. Bullish or bearish divergent formations.

The daily gold chart: a symmetrical triangle labelled extrinsically bullish, forming on a strong accumulation zone tested by multiple bottoms, beside a downtrend line acting as support, all inside a larger consolidation that contains smaller distributions and accumulations.

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Figure 4.12 A neutral triangle made extrinsically bullish.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

All five factors are outside the pattern. Figure 4.12, the daily gold chart, shows the second and the fourth at work on a neutral pattern. A symmetrical triangle forms on a strong accumulation zone that multiple bottoms have tested, beside a downtrend line acting as support, and so it is labelled extrinsically bullish. The trend phase follows after a short return move to the triangle's breakout barrier. The whole of it sits inside a larger consolidation that contains smaller distributions and accumulations. Divergence and cycles, the fifth and third factors, come back in Sections 5 and 9.

2.5 When the biases agree

Three rules of thumb follow, and the book's instruction for judging the reliability of a reversal or a continuation is to look for agreement. When intrinsic and extrinsic bias agree, a reversal at a top or bottom is more reliable. When the intrinsic bias agrees with the trend sentiment, a continuation is more likely. **Any disagreement means the reversal or continuation may be inherently weak.** Trend sentiment is the book's name for the direction of the trend, and it is used from here on.

A continuation pattern's bias should therefore agree with the trend sentiment. Bullish with respect to an uptrend, the book lists bullish pennants and flags, ascending triangles, inverted head and shoulders, rounding bottoms, cup and handles and falling wedges. Bearish with respect to a downtrend, it lists bearish pennants and flags, descending triangles, standard head and shoulders and rising wedges.

A 15 minute EURUSD chart: a distribution, a downside breakout, then a bear flag labelled intrinsically bearish and as continuation, with prices consolidating in a diagonally constrained manner before the trend resumes.

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Figure 4.10 A bear flag as continuation in a downtrend.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.10 is a bear flag on a 15 minute EURUSD chart. After a distribution and a downside breakout, prices consolidate in a diagonally constrained manner, and the flag is labelled intrinsically bearish and as continuation. Its bias agrees with the bearish trend sentiment, which greatly increases the potential for a reliable follow through. The flag is confirmed once price violates the pattern.

A 15 minute EURUSD chart with an ascending triangle, a flat top and a rising lower line, forming partway up an uptrend and labelled as continuation in a trend phase.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.13. An ascending triangle.

A 4 hour EURUSD chart with a rounding bottom, or cup and handle, forming partway up an uptrend and labelled as continuation in a trend phase.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.14. A rounding bottom, or cup and handle.

Figures 4.13 to 4.14 Two bullish patterns as continuation in an uptrend.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The book gives the bullish case in one sentence and two figures: two bullish patterns acting as continuation in trend phases.

Figure 4.13 is an ascending triangle, a flat top over a rising lower line, forming partway up an uptrend on a 15 minute EURUSD chart: a bullish pattern in a bullish trend. Figure 4.14 is a rounding bottom, or cup and handle, partway up an uptrend on a 4 hour chart. It is the only place the chapter draws a rounding bottom, so trace the cup, then the line across its rim.

2.6 Reversal patterns disagree with the trend

As a general guide, intrinsic bias and trend sentiment disagree for reversal patterns. A standard head and shoulders, intrinsically bearish, ends an uptrend as distribution. An inverted one, intrinsically bullish, ends a downtrend as accumulation.

A 15 minute EURUSD chart with a head and shoulders boxed as a consolidation range and labelled as distribution, a breakdown through its neckline, and a new consolidation forming at the projected target below.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.7 A head and shoulders as distribution.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

In Figure 4.7 a head and shoulders is boxed as a consolidation range and labelled as distribution. Price breaks down through its neckline, and at the projected target below a new consolidation forms. That target is the minimum price objective: a 1:1 projection of the pattern's height from the neckline breakout. It is the minimum measuring objective of Section 1 again, measured on a pattern and not on a range.

The book's summary diagram, Phase-Based Chart Patterns: three lists, accumulation based, distribution based and continuation based, with the first two marked as reversal patterns whose intrinsic and trend sentiment disagree, and the third as patterns whose intrinsic and trend sentiment should agree.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.5 The book's summary of the section: Phase-Based Chart Patterns.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.5 is the whole section on one page: three lists, accumulation based, distribution based and continuation based. Its type is small, so the first two lists are written out here.

Figure 4.5: the reversal patterns, by phase

- **Accumulation based, in downtrends:** inverted head and shoulders, cup and handles, rounding bottoms, falling wedges, ascending triangles, and double, triple and multiple bottoms.
- **Distribution based, in uptrends:** standard head and shoulders, rounding tops, rising wedges, descending triangles, and double, triple and multiple tops.
- Both lists add broadening, island and diamond formations. V bottoms and V tops are reversals too.

Bullish in a downtrend, bearish in an uptrend: in both lists the bias and the trend disagree, and the figure marks them as reversal patterns for that reason. Its third box is the continuation list, patterns whose intrinsic and trend sentiment should agree, which matches the two lists in 2.5.

2.7 Neutral patterns borrow their bias

A neutral pattern has no lean of its own, so its surroundings supply one. A neutral formation's extrinsic bias is derived from the trend sentiment: a symmetrical triangle is extrinsically bullish in an uptrend and extrinsically bearish in a downtrend. Broadening, diamond and island formations are reversals, so their bias opposes the trend.

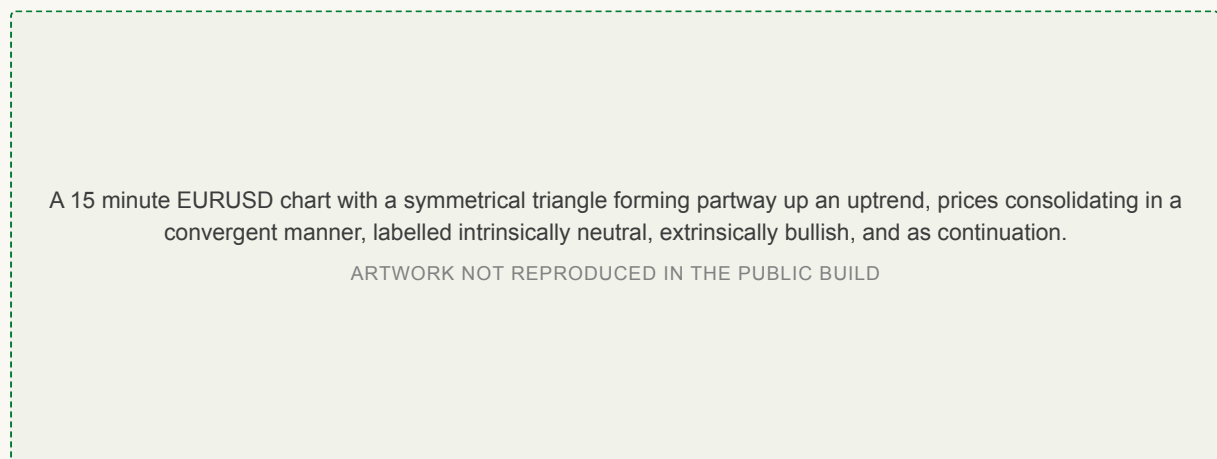


Figure 4.9 A neutral triangle borrowing the uptrend's bias.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.9 shows a symmetrical triangle forming partway up an uptrend, prices consolidating in a convergent manner, labelled intrinsically neutral, extrinsically bullish, and as continuation. The preexisting uptrend, in the book's word, imbues the triangle with extrinsic bullishness.

The book contradicts itself on broadening formations. Its list of patterns that are bullish with respect to an uptrend includes them, which files them with the trend. Its next paragraph says broadening, diamond and island formations are regarded as reversal formations, and its own Figure 4.5 puts them on both reversal lists. These notes teach the paragraph and the figure, and say plainly that the list disagrees.

2.8 Probabilities, never certainties

The daily iShares Silver Trust chart: two ascending triangles as accumulation, the second labelled intrinsically and extrinsically bullish, a gap up into the trend phase, then a new ascending triangle that breaks down unexpectedly, with volume declining into each breakout.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.11 Two triangles that worked, and one that failed.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.11, the daily iShares Silver Trust chart, is the book's warning. Two ascending triangles form the accumulation. The second forms inside the clear support the first one made, so it is labelled intrinsically and extrinsically bullish, and price breaks out on a large upside gap into the trend phase. After the trend a third ascending triangle forms, and then breaks down instead. Trend sentiment and intrinsic bias agreed perfectly, and the pattern still failed. **Technical forecasting is based on probabilities, and never on certainties.**

Failed patterns often lead to a more powerful reaction. Had the third triangle formed at a significant historical peak, the book adds, it would have been extrinsically bearish. Notice also the volume panel: volume declines as each breakout approaches, which is where Section 3 begins.

CHECK YOURSELF

A symmetrical triangle forms in the middle of a downtrend. What is its intrinsic bias, what is its extrinsic bias, and where does the second one come from?

Section 3

Volume and Open Interest Interpretation of Market Phase

What volume normally does in each phase, why it falls before a phase changes, what a trend that loses volume is saying, where open interest adds evidence, and what heavy volume marks. This is the book's section 4.3.

3.1 Volume falls before the phase changes

Each phase has its typical volume action: down through a consolidation, and up through a trend. Volume tends to decline to relatively low levels just before a phase transition, so it can be used as a timing indicator for potential phase transitions. **Very low volume says a consolidation may be ending. It does not say which way.** The book's summary calls a phase transition a market regime change, which is the phrase in the chapter's third learning objective.

Volume should rise through a trend. If an uptrend or downtrend is inherently weak, volume falls off during the trend phase. That is a sign of trend exhaustion, and it signals a potentially earlier onset of a distribution or accumulation phase: the trend ends sooner than its start suggested.

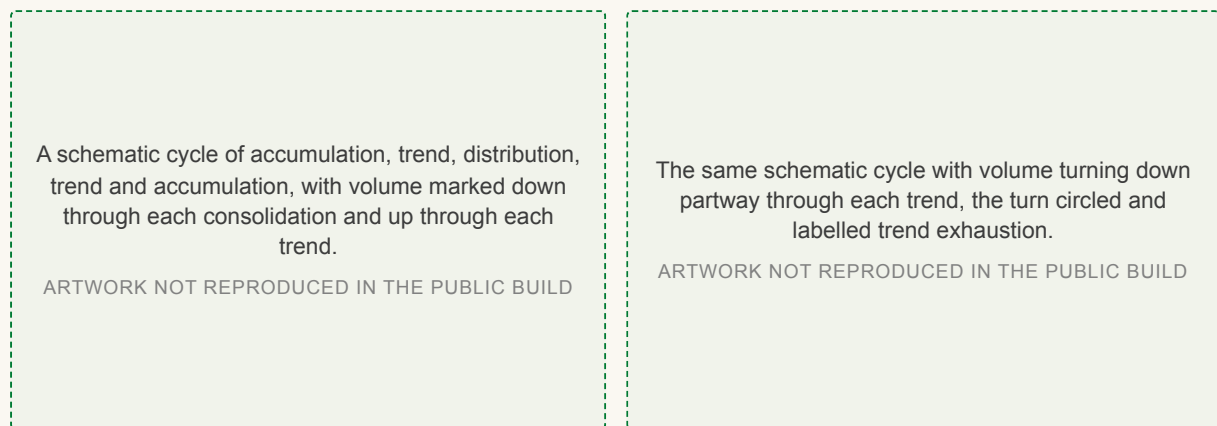


Figure 4.15. Typical volume action.

Figure 4.16. Trend exhaustion.

Figures 4.15 to 4.16 Volume through the cycle, in the normal case and in a weak trend.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.15 is the normal case as a schematic cycle of accumulation, trend, distribution, trend and accumulation, with volume marked down through each consolidation and up through each trend. Figure 4.16 is the same cycle with volume turning down partway through each trend, the turn circled and labelled trend exhaustion. Compare the two trends: in the first, volume rises all the way through.

3.2 Open interest: watch for disagreement

For the most part, open interest action correlates fairly closely with volume action. It is when the two diverge significantly that open interest adds evidence. **If open interest increases during a consolidation instead of decreasing, the breakout and the trend after it tend to be much more rapid and extended.**

Open interest is used here and has still not been defined. Chapter 3 listed it as transaction data, and the book teaches it in its Chapter 6. These notes do not borrow a definition in the meantime.

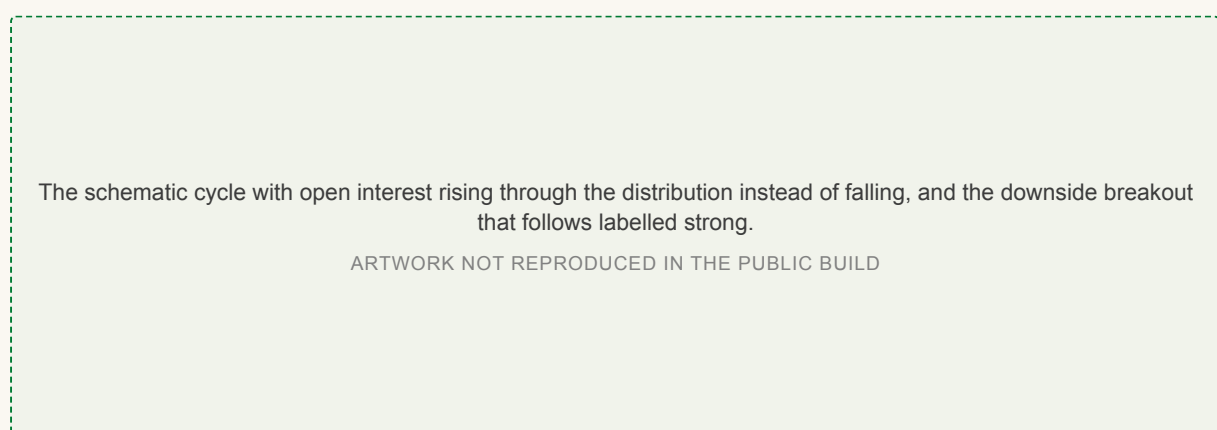


Figure 4.17 Open interest rising through a distribution.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.17 shows the schematic cycle with open interest rising through the distribution instead of falling, and the downside breakout that follows labelled strong.

3.3 Volume and phase on a real chart

An intraday Isis Pharmaceuticals chart: a long distribution phase broken by a downside gap, a trend phase curving down into an accumulation phase, and a volume panel labelled bullish volume divergence under the trend and again under the accumulation.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.18 Distribution, trend and accumulation, with volume.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.18, an intraday Isis Pharmaceuticals chart, puts the three schematics on one stock. An extended distribution is violated by a downside gap, on high volume. The trend phase follows and passes into an accumulation, and volume gradually decreases as the accumulation progresses. **Falling volume signals a potentially imminent breakout, to the upside or the downside.** The figure labels its volume panel bullish volume divergence, under the trend and again under the accumulation. Divergence is the subject of Section 5.

3.4 Heavy volume marks a top or a bottom

Higher than normal volume at the start of an accumulation or distribution strongly indicates that a bottom or top has formed. **Price levels with historically high volume are therefore potential levels of accumulation or distribution.**

An hourly Facebook chart with horizontal volume bars drawn against price levels: the largest coincide with a distribution zone and an accumulation zone, with order flow marked seeking out additional liquidity at the edges of each.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.19 Price-based volume at the two consolidation zones.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

In Figure 4.19, an hourly Facebook chart, horizontal volume bars are drawn against price levels, and the largest coincide with a distribution zone and an accumulation zone. The figure also marks order flow seeking out additional liquidity at the edges of each zone: the market tests the boundaries of a consolidation. This is price-based volume, as opposed to time-based volume, and the chapter uses it without explaining it.

So volume has two uses in this section. Falling volume times the end of a consolidation, and heavy volume marks where one began.

CHECK YOURSELF

Volume has been falling for weeks inside a range. What does that tell you about the range, and what does it not tell you?

Section 4

Moving Average Interpretation of Market Phase

How one moving average behaves in a trend and in a consolidation, and what several of them do together. This is the book's section 4.4.

4.1 One moving average across the phases

Market phase can also be identified by the behavior of moving averages. During a trending phase a moving average shows the minimum incidence of whipsaws. During a consolidation, whipsaw action increases significantly. The cause is the flattening-out effect of averages in a sideways or ranging market.

One simple moving average under a rising price, labelled as containing the trend with fewer whipsaws, then flattening inside a boxed consolidation where price crosses it again and again, labelled whipsaws during the consolidation phase.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.20 One moving average in a trend and in a range.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.20 shows one simple moving average under a rising price, labelled as containing the trend with fewer whipsaws. The average then flattens inside a boxed consolidation, where price crosses it again and again, labelled whipsaws during the consolidation phase. **The book uses the word whipsaw without defining it.** In the figure the whipsaws are the places where price crosses back and forth over the flat average. That reading comes from the figure's own labels; moving averages, and the definition, are the book's Chapter 11.

4.2 Several averages: spread in a trend, tangled in a range

In a trend, multiple moving average lines start to diverge from each other, and the greater the divergence, the stronger the ensuing trend. When the lines begin to converge, that is an early indication of a potential consolidation. **Overlapping lines mean a consolidation, and diverging lines mean a strong trend in place.** That is the answer to the review question on using moving averages to gauge market phase.

A daily chart with four moving averages, 20, 50, 100 and 200 day: the lines overlap through an accumulation phase, fan apart through the trend phase, where they also act as support, and overlap again in the consolidation that follows.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.21 Four moving averages through the phases.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.21 is a daily chart with four moving averages, of 20, 50, 100 and 200 days. The lines overlap through an accumulation phase, fan apart through the trend phase, and overlap again in the consolidation that follows. The figure also labels the averages acting as support during the trend. The book's text does not discuss that here.

Section 5

Divergence and Momentum Interpretation of Market Phase

What divergence is used for in phase analysis, how it joins with a pattern and a cycle, and why momentum turning before price is an early sign. This is the book's section 4.5.

5.1 Divergence tracks momentum

Divergence may be used to track the changes in price momentum. That makes it extremely useful for identifying potential consolidations, and the strength of a trend. Once divergence is observed, look for a slowing of the trend.

The chapter uses divergence, MACD and RSI without defining any of them. Divergence is taught in the book's Chapter 9. What follows describes what each figure shows, in the book's own labels, and stops there.

In Figure 4.22, an hourly EURUSD chart, price makes a higher peak inside a distribution while the MACD below it makes a lower one. The figure labels this standard bearish divergence, and momentum turning before price. Divergence shows momentum slowing, and a slowing trend is a consolidation forming, which is the answer to the review question on divergence.

An hourly EURUSD chart: price makes a higher peak inside a distribution while the MACD below it makes a lower one, labelled standard bearish divergence and momentum turning before price.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.22 Standard bearish divergence in a distribution.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

5.2 Bullish divergence supports an accumulation

The daily 3M chart: an accumulation in the form of an inverted head and shoulders, with standard bullish divergence marked on the MACD below it and vertical lines marking consistent cycle lows at the shoulders and the head.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.23 A pattern, a divergence and a cycle in agreement.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.23 is the daily 3M chart. The accumulation takes the form of an inverted head and shoulders pattern, and it is further supported by standard bullish divergence on the MACD. The left shoulder, the head and the right shoulder coincide with the projected cycle lows, which the figure draws as vertical lines. **This bullish confluence within the accumulation greatly increases the probability of a strong upside move.** Confluence is the book's word for several kinds of evidence agreeing: here a pattern, a divergence and a cycle. The figure also labels a reverse bullish and a standard bearish divergence, and the text discusses neither.

5.3 Momentum turns before price

A 5 minute GBPUSD chart: a downtrend into an accumulation, with bullish divergence marked on volume, on the MACD and on the RSI inside the accumulation, and a new trend rising out of it.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.24 Bullish divergence on volume, the MACD and the RSI.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.24 is a 5 minute GBPUSD chart of a downtrend passing into an accumulation, with a new trend rising out of it. Volume, the MACD and the RSI all show bullish divergence with price in the accumulation, and the MACD and the RSI are both momentum-based indicators. Price is still falling inside the box while both indicators are already rising. **We see momentum turning before price.** At the tail end of a trend, a sign that a consolidation may be forming helps a trader anticipate an early entry.

The book gives one sentence of method, and it is about volume: for bullish divergence in volume, falling prices must be accompanied by decreasing volume.

CHECK YOURSELF

Price makes a higher peak while a momentum indicator makes a lower one. What should you look for in the trend next, and which phase may be forming?

Section 6

Sentiment Interpretation of Market Phase

Why sentiment reads best at market extremes, the thirteen indicators the book names, and what six of them read at a top and at a bottom. This is the book's section 4.6.

6.1 Sentiment is clearest at the extremes

Market participants swing between fear, greed and hope, responding to news and to each other. Identifiable patterns repeat fairly consistently, especially at market extremes. Sentiment indicators track participants' aggregate sentiment, emotions and psychology. **They are most accurate at extremes of optimism or pessimism.**

Section 1 already described that crowd. The uninformed were extremely bullish at the distribution and extremely bearish at the accumulation. Sentiment indicators measure them, so they read best exactly where the crowd is most one sided.

Chart Q boxes the top and the bottom of one market swing. At the top the crowd is extremely bullish, at the bottom extremely bearish, and those two extremes are where sentiment indicators are most accurate.

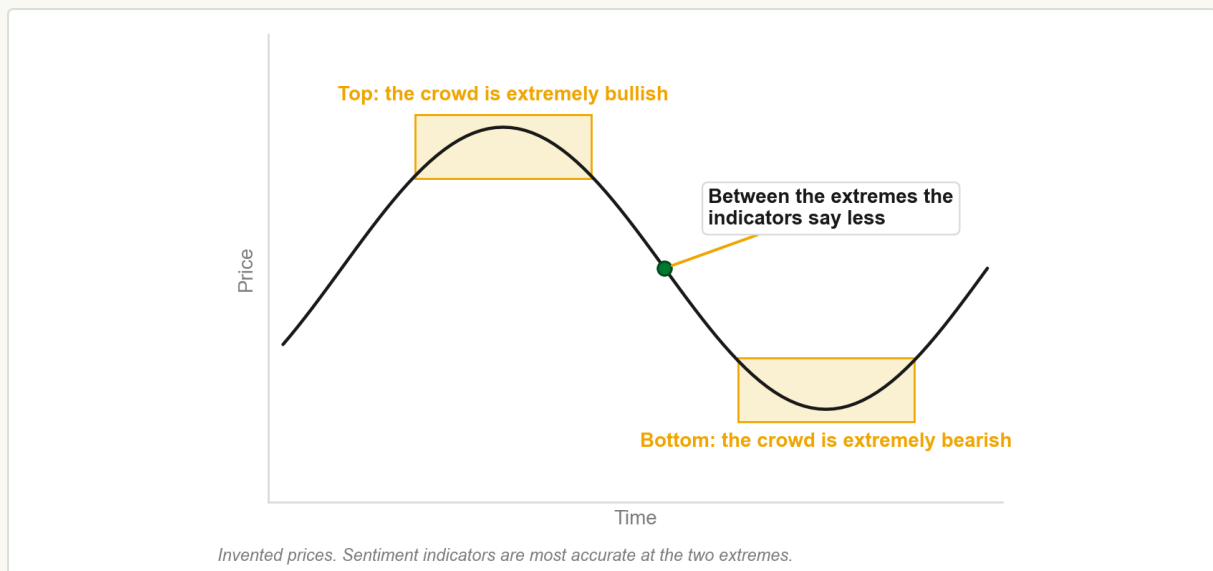


Chart Q Where sentiment indicators are most accurate.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

6.2 Thirteen indicators, by name

The book gives thirteen sentiment indicators as examples: put-call ratios, Arm's Index, odd lot sales, margin debt, the COT report, the VIX, the short interest ratio, the cash asset ratio, the advance-decline line, new highs minus new lows, up volume minus down volume, Market Vane reports, and the Bullish Sentiment Index. **It explains none of them in this chapter, and these notes do not either.** The book's Chapter 23 is the detailed discussion. How any one of them is built is not examined from this chapter.

6.3 What the indicators read at a top and at a bottom

At a distribution the put/call ratio and the VIX are low, but starting to increase. A/D, margin debt, new highs minus new lows and the Bullish Percent Index are high, but starting to decrease. At an accumulation every one of those readings is reversed. **Each reading is at an extreme and has just started to turn.** This is the answer to the review question that asks which sentiment indicators best describe distributions.

A price curve from a distribution at the top to an accumulation at the bottom, with six sentiment readings listed beside each: put/call ratio, A/D, VIX, margin debt, new highs minus new lows and the Bullish Percent Index.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.25 Six sentiment readings at a top and at a bottom.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.25 draws a price curve from a distribution at the top to an accumulation at the bottom and lists the six readings beside each. Two start low at a top and four start high. At a bottom, swap the words high and low.

One loose end. The figure says Bullish Percent Index, and the list of thirteen says Bullish Sentiment Index. The book does not say whether they are the same.

CHECK YOURSELF

Cover Figure 4.25 and write out the six readings for an accumulation. Which two are high, and which way has each started to turn?

Section 7

Sakata's Interpretation of Market Phase

A Japanese model that divides the market into five phases, the five names and what each stands for, and how the five fit inside Dow's three. This is the book's section 4.7.

7.1 Five phases, not three

Dow's model has three phases. The Japanese way of reading the market divides it into five: three bottoms, three thrusts with a correction after each, and three tops.

Sakata's Five Methods. A unique Japanese approach for interpreting and understanding the markets, which originated with Munehisa Honma. The market is divided into five distinct phases, as opposed to the conventional three-phase model by Dow.

Two panels side by side: Sakata's five market phases, with San Sen at the bottom, San Pei, San Poh and San Ku through the rise and San Zan at the top, beside Elliott's 5-3 wave structure.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.26 Sakata's five phases, beside Elliott's waves.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.26 sets two panels side by side. On the left are Sakata's five market phases: San Sen at the bottom, San Pei, San Poh and San Ku through the rise, and San Zan at the top. On the right is Elliott's 5-3 wave structure, which is Section 8. The book says the two are very similar in appearance. Munehisa Honma's name is all the history the book gives, and these notes add none.

7.2 The five methods, by name

- **San Zan, three mountains.** The Western triple top. The distribution phase.
- **San Sen, three rivers.** The Western triple bottom. The accumulation phase.
- **San Ku, three gaps.** The breakaway, runaway and exhaustion gaps of the trend.
- **San Pei, three thrusts.** The three trending phases between the two.
- **San Poh, three methods.** The three corrections, one after each thrust.

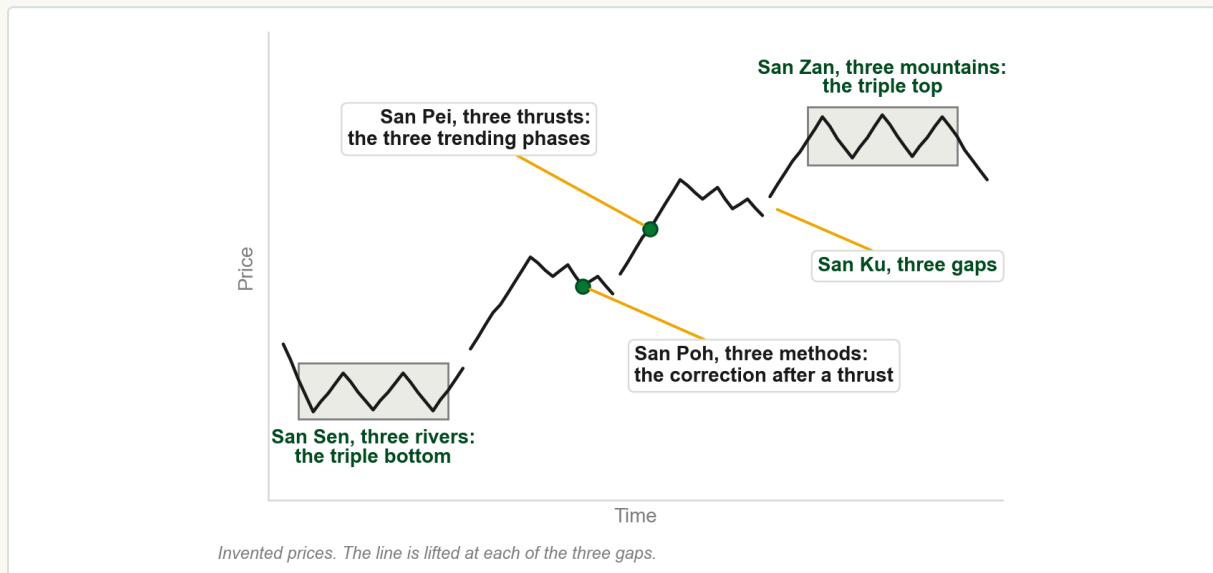


Chart R The five methods on one line.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

The book glosses every name as three of something, which is the handle for remembering all five. Two of the five are consolidations, and three describe the trend between them. Chart R puts all five on one invented line, each with the book's English. The three gaps are among the four that Chapter 3 named.

7.3 Five phases inside three

Dow's three phases may seem inconsistent with Sakata's five, and with Elliott's waves. The book's answer is that simple chart patterns show the models need not be viewed as distinct: a triple bottom, then thrusts with bullish flags and gaps between them, then a triple top. **It calls this a plausible resolution between Eastern and Western approaches, and its summary calls the two approaches complementary.**

Dow's three-phase market drawn with Sakata's pieces: a triple bottom as the accumulation phase, a rising channel of thrusts, bullish flags and the breakaway, runaway and exhaustion gaps as the trend phase, and a triple top as the distribution phase.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.27 Dow's three phases drawn with Sakata's pieces.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.27 draws Dow's three-phase market with Sakata's pieces: a triple bottom as the accumulation phase, a rising channel of thrusts, bullish flags and the breakaway, runaway and exhaustion gaps as the trend phase, and a triple top as the distribution phase. This is the chapter's sixth learning objective, comparing Eastern and Western approaches, and the comparison comes out as one fitting inside the other.

CHECK YOURSELF

Without looking back, give the English for each of the five names, and say which two are consolidations.

Section 8

Elliott's Interpretation of Market Phase

Elliott's five waves up and three waves down mapped onto the phases, and a wave count on a real chart. This is the book's section 4.8.

8.1 Five waves up, three waves down

Elliott described markets as an impulsive five-wave up, then a corrective three-wave down. **The trending phase is impulse waves 1, 3 and 5, and the accumulation phase is corrective waves 2 and 4.** Sakata's San Pei closely resembles waves 1, 3 and 5, and San Poh somewhat resembles waves a, b and c. Elliott's waves have specific rules; Sakata's model is less rigid.

Elliott's wave structure: five numbered motive waves rising, with waves 1, 3 and 5 marked impulsive, then three lettered corrective waves a, b and c falling. Eight waves of one degree.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.28 Elliott's wave structure.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.28 is the structure: five numbered motive waves rising, with waves 1, 3 and 5 marked impulsive, then three lettered corrective waves a, b and c falling. That is eight waves of one degree.

The book gives the abc correction to both consolidations. It says the accumulation phase includes the abc correction in most cases, and in the next sentence that distribution is usually the abc correction. These notes record both and do not pick one.

8.2 The waves on the gold chart

The weekly gold chart from 2000 to 2012 with an Elliott wave count on log scaling: an accumulation phase, two trend phases with a consolidation between them, a potential distribution phase at the top, and volume marked falling across three consolidations.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 4.29 An Elliott wave count on weekly gold.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 4.29 is the weekly gold chart from 2000 to 2012 with an Elliott wave count, on log scaling. Primary waves 1 and 2 sit in the accumulation, and the powerful wave 3 in the trend. Volume gradually decreases over each significant consolidation, at the three places the figure numbers, which is Section 3 again. Volume peaks around the top of wave 5, within a new area of consolidation that the figure marks as a potential distribution phase. The book describes that consolidation at the top as an inverted head and shoulders, and these notes report it as the book states it.

The book adds that Fibonacci price and time projections help gauge where each wave ends. Fibonacci projection is not taught in this chapter, and Elliott's rules are the book's Chapter 18.

Section 9

Cycle Analysis Interpretation of Market Phase

How a formation's place on a cycle decides whether it is a reversal or a continuation, shown on one neutral pattern in three roles, and a list of what this chapter uses without teaching. This is the book's section 4.9.

9.1 Where in the cycle decides what the pattern is

Whether a formation reverses or continues depends on where in the cycle it unfolds. Between cycle extremes, formations tend to be continuations. At cycle peaks they tend to be distributions, and at cycle troughs, accumulations. **If the intrinsic and extrinsic biases agree as well, the formation is potentially more reliable.** All of this is for formations on the same cycle degree, and it is the answer to the review question on using a cycle to gauge market phase.

It is also extrinsic bias again: the phase of an underlying market cycle was the third of the five factors in Section 2. Once a cycle period is identified, the book says, placing the formation on it is fairly simple. How the period is found is the book's Chapter 20.

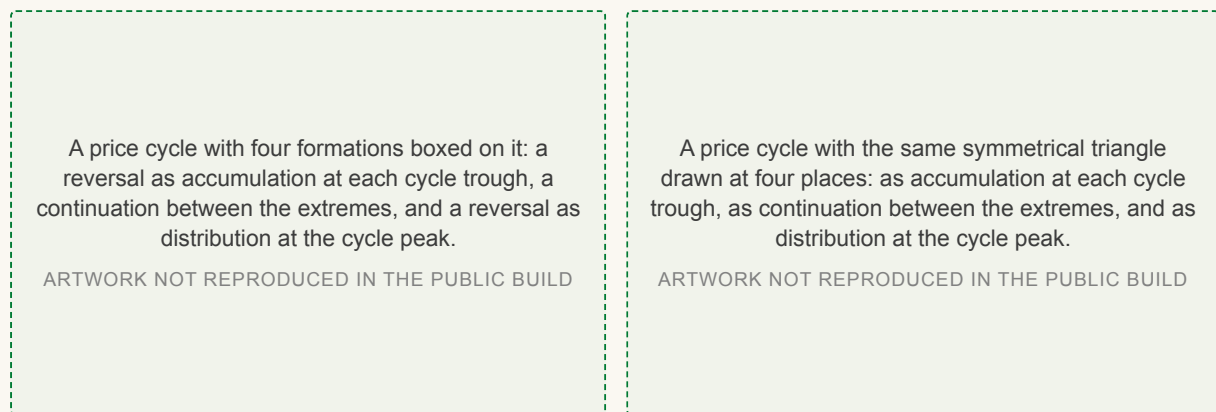


Figure 4.30. Formations on a cycle.

Figure 4.31. One triangle, three roles.

Figures 4.30 to 4.31 Where a formation sits on the cycle, and what that makes it.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.30 boxes four formations on a price cycle: a reversal as accumulation at each cycle trough, a continuation between the extremes, and a reversal as distribution at the cycle peak. Figure 4.31 does the same with a single pattern, the symmetrical triangle, drawn at four places. The shape is the same each time, and only its place on the cycle differs.

9.2 One triangle, three roles

The intrinsically neutral symmetrical triangle takes three distinct roles: accumulation at the bottom of the cycle, continuation between the top and the bottom, and distribution at the top. It adopts a bullish extrinsic bias at cycle troughs and a bearish one at cycle peaks. Between the extremes it adopts the preexisting trend sentiment. Section 2 said that neutral patterns borrow their bias, and here the lender is the cycle and not the trend.

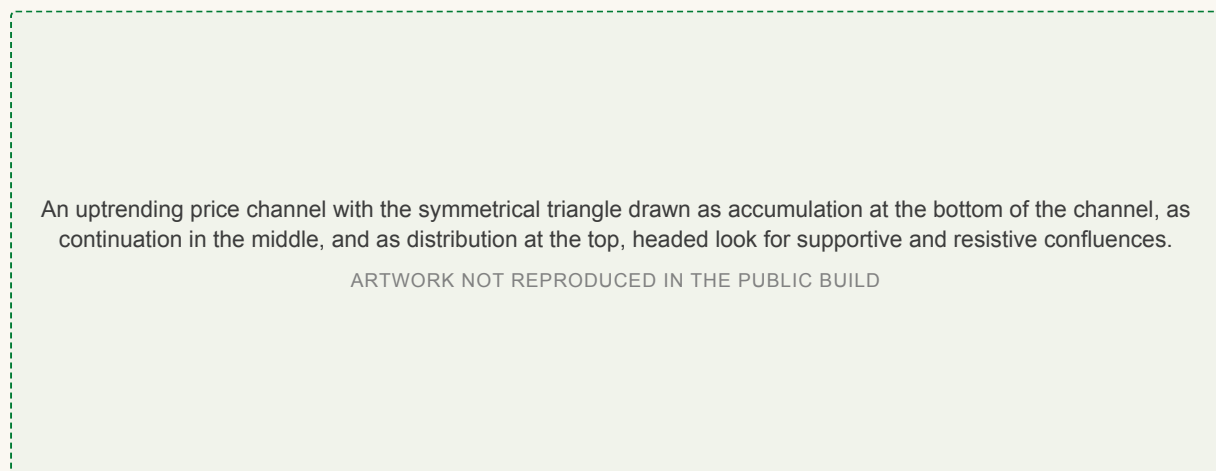


Figure 4.32 The same triangle in a price channel.

Reproduced from Lim, *The Handbook of Technical Analysis* (Wiley, 2016)

Figure 4.32 moves the triangle into an uptrending price channel: accumulation at the bottom of the channel, continuation in the middle, and distribution at the top. Had the pattern been an inverted head and shoulders, the book says, a reversal at the bottom or a continuation in the middle would be more probable. That pattern is intrinsically bullish, so at the bottom of the channel both biases would agree.

The figure is headed with an instruction to look for supportive and resistive confluences. **Confluences with other overlay barriers, such as Fibonacci levels, give more reliable forecasts of where price may react.** Fibonacci levels are named here and not taught.

CHECK YOURSELF

A symmetrical triangle forms at a cycle peak. What role does it take, and what extrinsic bias does it adopt?

9.3 What this chapter uses and does not teach

Six of the nine sections borrow a tool that a later chapter of the book teaches. Each was named where it appeared, and they are collected here.

- The shapes of the chart patterns: Chapter 13.
- Open interest: Chapter 6. Price-based volume is also used and not explained.
- Moving averages and whipsaws: Chapter 11.
- Divergence, MACD and RSI: Chapter 9.
- The thirteen sentiment indicators: Chapter 23.
- Elliott's rules: Chapter 18. Fibonacci projections are also used and not explained.
- How a cycle period is found: Chapter 20.
- The learning objectives promise the inertial characteristics of each phase. The text never uses the word.

From this chapter, only what each tool says about market phase is examinable. Do not fill these gaps from outside the book. Each comes back in its own chapter.

Three more places are worth remembering, because the book says two different things or leaves a question open. Broadening formations are filed two ways, in 2.7. The abc correction is given to both consolidations, in 8.1. And the Bullish Percent Index of Figure 4.25 may or may not be the Bullish Sentiment Index of the list, in 6.3.

Chapter 4 in five sentences

Summary

1. Markets consolidate or trend; a consolidation is accumulation or distribution.
2. Which one it was is known only afterwards, so the analyst gathers evidence.
3. Chart patterns are evidence: their intrinsic and extrinsic bias should agree.
4. Volume, moving averages, divergence, sentiment and cycles each add evidence.
5. Sakata's five phases and Elliott's waves fit inside Dow's three.

Review questions

These are the chapter's own review questions. Every one is answered somewhere in the nine sections above, and they are the shape the quiz takes.

1. Define the term consolidation, and explain why most chart patterns are consolidation patterns.
2. List eight chart patterns that are intrinsically bullish.
3. Describe how you would determine if accumulation is taking place.
4. Which sentiment indicators best describe distributions?
5. How would you use moving averages, and a cycle, to gauge market phase?
6. How do you use divergence to identify potential consolidations?

Key terms

Every term this chapter defines, against the subsection that defines it.

Consolidation phase	1.1	Intrinsic bias	2.3
Minimum measuring objective	1.8	Extrinsic bias	2.4
Consolidation pattern	2.2	Sakata's Five Methods	7.1

Sources

- Lim, M. (2016). The Handbook of Technical Analysis. Wiley. Chapter 4, and the source of every figure here.